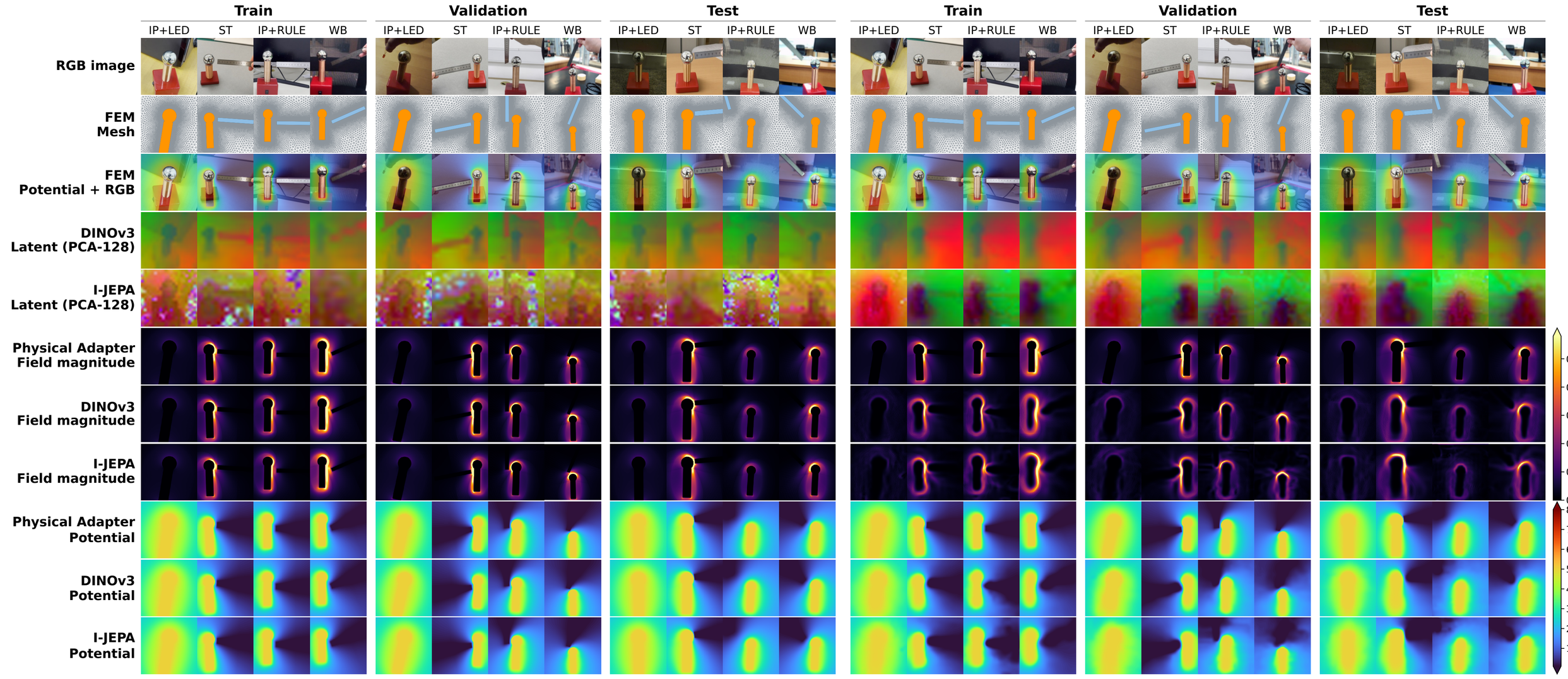

(a) Frozen-encoder residual adapters
(b) Full fine-tuning (Full FT)
Train
Validation
Test
IP+LED
ST
IP+RULE
WB
RGB image
FEM Mesh
FEM Potential + RGB
DINOv3 Latent (PCA-128)
I-JEPA Latent (PCA-128)
Physical Adapter Field magnitude
DINOv3 Field magnitude
I-JEPA Field magnitude
Physical Adapter Potential
DINOv3 Potential
I-JEPA Potential
$|E_p|$ (eq. V/pixel)
0.25
0.20
0.15
0.10
0.05
0.00
LED-equivalent V
8
7
6
5
4
3
2
1